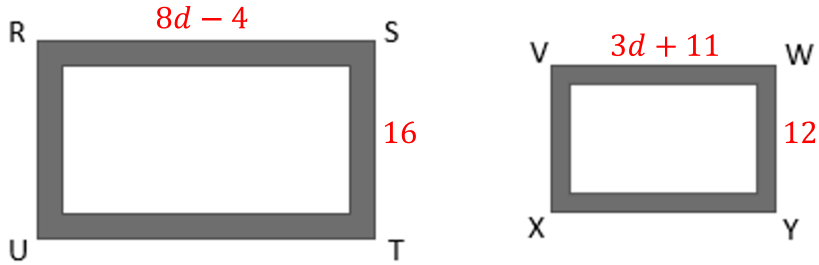
Geometry
Unit 6
Lesson 12 Homework

Name___Answer Key_______
Date_____________________
Period___________________

Hillary is custom building frames for a set of family photos. She has designed two similar rectangular frames, $RSTU$ and $VWYX$. The diagram shows the two different sized frames, where $WY = 12$ inches (in.), $VW = (3d + 11)$ in., $ST = 16$ in., and $RS = (8d - 4)$ in.



1. What is the value of d?

$\frac{8d-4}{3d+11} = \frac{16}{12}$ $\quad$ $12(8d - 4) = 16(3d + 11)$ $\quad$ $\boxed{d = 4\frac{2}{3}}$

$96d - 48 = 48d + 176$

$48d = 224$

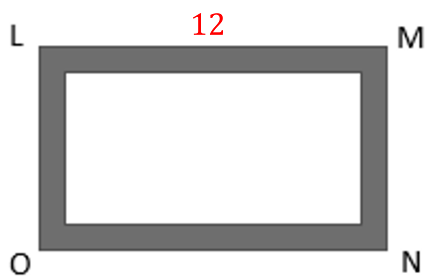
2. What is the area of $RSTU$? Include units.

$RS = 8\left(4\frac{2}{3}\right) - 4 = 33\frac{1}{3}$ $\quad$ Area $= \left(33\frac{1}{3}\right)(16) = \boxed{533\frac{1}{3}\text{ in}^2}$

3. What is the perimeter of $VWYX$? Include units.

$VW = 3\left(4\frac{2}{3}\right) + 11 = 25$ $\quad$ $P = 2(25) + 2(12) = \boxed{74\text{ in}}$

Hillary created a third similar rectangular frame, $LMNO$, shown below, where $LM = 12$ in.



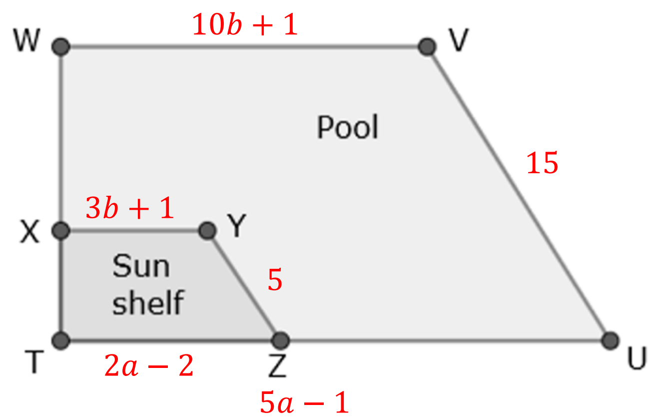
4. What is the length of $\overline{LO}$? Include units.

$\frac{25}{12} = \frac{12}{LO}$ $\quad$ $LO = \frac{(12)(12)}{25} = \boxed{5.76\text{ in}}$

5. What is the perimeter, in inches, of $LMNO$?

$P = 2(12) + 2(5.76) = \boxed{35.52\text{ in}}$

Laurel and her family are having a pool built in the backyard. The pool, $TUVW$, is shaped like a trapezoid, with a smaller similar trapezoid, $TZYX$, for a sun shelf. A diagram of the pool is shown, where $TU = (5a - 1)$ feet (ft), $TZ = (2a - 2)$ ft, $ZY = 5$ ft, $UV = 15$ ft, $VW = (10b + 1)$ ft, and $YX = (3b + 1)$ ft.



6. Determine the value of each variable:

$a =$ __5____
$b =$ __2____

$\frac{15}{5} = \frac{5a-1}{2a-2}$
$15(2a - 2) = 5(5a - 1)$
$30a - 30 = 25a - 5$
$5a = 25$
$a = 5$

$\frac{15}{5} = \frac{10b+1}{3b+1}$
$15(3b + 1) = 5(10b + 1)$
$45b + 15 = 50b + 5$
$5b = 10$
$b = 2$

7. Determine the value of each side:

$WV =$ __21____ $\quad WV = 10(2) + 1 = 21$
$TZ =$ __8____ $\quad TZ = 2(5) - 2 = 8$

8. If $WV = WT$, what is the perimeter, in feet, of the pool?

$P = 21 + 21 + 15 + [5(5) - 1] = 81$ ft

Some measurements of angles of $TUVW$ and $TZYX$ are given.

$m\angle VUZ = (12c - 6)°$, $m\angle YZT = (10c + 5)°$, $m\angle XTZ = 90°$,
$m\angle TXY = (4d + 10)°$, $m\angle TWV = (5d - 10)°$

9. Determine the value of each variable:

$c =$ __$\frac{11}{2}$____
$d =$ __20____

$12c - 6 = 10c + 5$
$2c = 11$
$c = \frac{11}{2}$

$4d + 10 = 5d - 10$
$d = 20$

10. What is the measure of $\angle XYZ$?

$10\left(\frac{11}{2}\right) + 5 = 55 + 5 = 60°$
$4(20) + 10 = 80 + 10 = 90°$

$60 + 90 + 90 = 240°$
$360° - 240° = 120°$

$\boxed{m\angle XYZ = 120°}$